Introduction to Algebra

Algebra is a branch of mathematics dealing with symbols and the rules for manipulating those symbols. In elementary algebra, those symbols (today written as Latin and Greek letters) represent quantities without fixed values, known as variables. Just as sentences describe relationships between specific words, in algebra, equations describe relationships between variables. The field of algebra evolved continuously over several millennia from ancient civilizations including Babylon and Egypt.

Variables and Expressions

A variable is a symbol used to represent an unknown or changeable value in a mathematical expression or equation. Common variable names are x, y, and z, but any letter can be used. An algebraic expression combines variables, numbers, and operation signs. For example, $3x + 5$ is an expression where x is a variable and 3 and 5 are constants. Evaluating an expression means substituting a value for each variable and then computing the result using the standard order of operations.

Linear Equations

A linear equation is an equation that forms a straight line when graphed on a coordinate plane. It contains only constants and variables raised to the first power. The standard form of a linear equation in one variable is $ax + b = c$, where a, b, and c are constants and x is the variable. Solving a linear equation means finding the value of the variable that makes the equation true. We isolate the variable by performing the same operation on both sides of the equation to maintain equality.

Inequalities

An inequality is a mathematical statement that compares two expressions using an inequality sign such as less than, greater than, less than or equal to, or greater than or equal to. Unlike equations, inequalities typically have infinitely many solutions, which are represented as a range of values. When solving an inequality, the same rules apply as for equations with one important exception: multiplying or dividing both sides by a negative number reverses the direction of the inequality sign.

Systems of Equations

A system of equations is a set of two or more equations that share the same variables. The solution to a system is the set of values that satisfy all equations simultaneously. There are three main methods for solving systems of linear equations: graphing, substitution, and elimination. Graphing involves plotting both lines and identifying the point of intersection. Substitution means solving one equation for a variable and plugging the result into the second equation. Elimination adds or subtracts equations to cancel a variable.

Polynomials

A polynomial is an expression consisting of variables and coefficients combined using addition, subtraction, and multiplication, with non-negative integer exponents. Polynomials are classified by their degree, which is the highest power of the variable. A monomial has one term, a binomial has two terms, and a trinomial has three terms. Adding and subtracting polynomials involves combining like terms, which are terms that have the same variable raised to the same power. Multiplying polynomials requires applying the distributive property repeatedly.

Factoring

Factoring is the process of writing a polynomial as a product of simpler polynomials or other mathematical objects. It is the inverse operation of expanding. Common factoring techniques include factoring out the greatest common factor, factoring trinomials into two binomials, difference of squares, and sum and difference of cubes. Factoring is useful for simplifying expressions, solving polynomial equations, and finding zeros of functions. The zero product property states that if a product of factors equals zero, then at least one factor must equal zero.